

WEEK -5

Healthcare Data Cleaning and Data Modelling

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Introduction

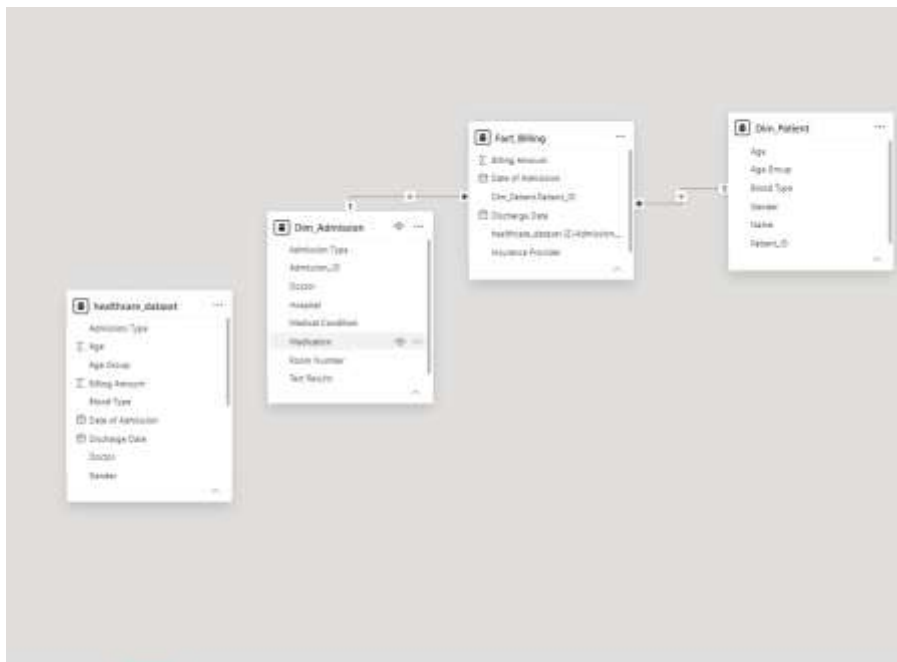
During Week 5 of my Power BI project, I worked on data modelling for the healthcare dataset.

The main objective was to organize the healthcare data into separate tables and establish relationships between them. This helped create a structured data model that can be used for further analysis and reporting.

During this week, I worked with the following tables:

- Dim_Admission
- Dim_Patient
- Billing

I also created the required relationships between these tables in Power BI Model view



2. Creating the Dim_Admission Table

Dim_Admission

I created the Dim_Admission table to store information related to patient admissions.

The table contains the following fields:

- Admission Type
- Admission ID
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number

Test Results

These fields provide information about the patient's admission, treatment and hospital details.

The Admission ID is used as an important field for connecting admission information with the billing data.

3. Creating the Dim_Patient Table

Dim_Patient

Next, I created the Dim_Patient table to store patient-related information.

The table contains:

- Age
- Age Group
- Blood Type
- Gender
- Name
- Patient ID

The Patient ID helps identify each patient and is also used to connect the patient information with the Billing table.

This table makes it easier to use patient information for analysis.

4. Creating the Billing Table

Billing Table

I created the Billing table to store billing and admission date information.

The Billing table contains:

- Admission-ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider
- Patient ID

The Billing Amount contains the amount related to the patient's billing.

The Date of Admission and Discharge Date can later be used for date-based analysis.

The Insurance Provider field provides information about the patient's insurance company.

The Patient ID connects the billing information with the Dim_Patient table.

5. Connecting the Tables

Creating Relationships

After creating the three tables, I connected them in the Model view of Power BI.

The tables were connected using the appropriate ID fields.

The main relationships are:

Dim_Admission → Billing

and

Dim_Patient → Billing

The relationship between Dim_Admission and Billing uses the admission-related ID.

The relationship between Dim_Patient and Billing uses Patient ID.

These relationships allow information from different tables to work together when creating Power BI reports.

6. Checking the Data Model

Model View

After creating the relationships, I checked the complete model in Power BI's Model view.

The final model showed:

Dim_Admission → Billing ← Dim_Patient

The relationship symbols were visible between the tables.

I checked that:

- The tables were connected.
- The correct fields were used for relationships.
- The Billing table was connected to both dimension tables.
- Patient information was connected through Patient ID.
- Admission information was connected through the admission ID.

7. Final Data Model

Completed Model

The final Week 5 data model contains three tables:

1. Dim_Admission

Contains admission and hospital-related information.

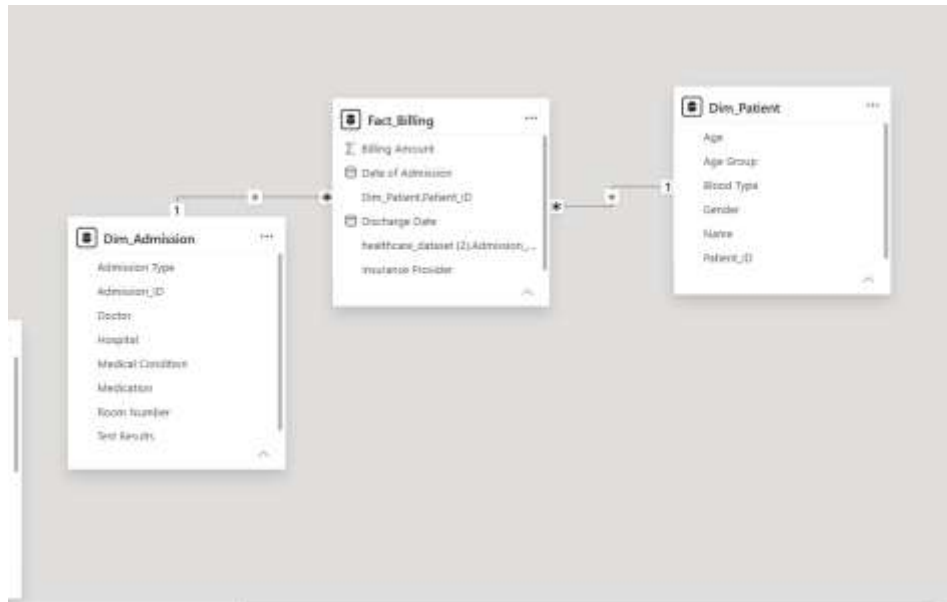
2. Dim_Patient

Contains patient-related information.

3. Billing

Contains billing, admission date, discharge date, insurance provider and patient ID information.

The completed model provides a structured way to work with the healthcare dataset.



8. Conclusion

In Week 5, I completed the data modelling part of my Power BI healthcare project.

I organized the dataset into three tables: Dim_Admission, Dim_Patient and Billing. I then created relationships between these tables using the required ID fields.

The final model is now organized and connected properly. This provides the base for continuing the Power BI project in the next week.